

Bayesian Parameter Estimation from Homodyne Quadrature Measurements: Gaussian and Non-Gaussian Examples

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Bayesian parameter estimation provides a natural framework for inferring physically meaningful parameters of quantum states directly from measurement data while retaining uncertainty information beyond a single best-fit estimate. In this report, I study its application to continuous-variable optical states reconstructed from homodyne quadrature measurements. Two representative models are considered: a Gaussian squeezed-thermal state, parametrized by the squeezing strength, squeezing angle, and thermal occupation, and a non-Gaussian vacuum–single-photon mixture described by a single mixing parameter. For each model, both simulated and experimental datasets are analyzed using Markov-chain Monte Carlo sampling of the posterior distribution. The inferred parameters, posterior correlations, and central credible intervals are examined as the number of homodyne measurements is increased. In the simulated cases, the posterior estimates approach the known ground-truth values while the uncertainty contracts approximately with the expected $N^{-1/2}$ scaling. The experimental cases are used to study posterior stabilization in the absence of independently known true parameters. This work is not intended to introduce a new estimation method, but rather to provide a compact and reproducible study of Bayesian parameter estimation for Gaussian and non-Gaussian quantum optical states, with emphasis on practical implementation, uncertainty quantification, and the interpretation of posterior convergence.

I. INTRODUCTION

Quantum optics has recently offered a promising platform for quantum computing and quantum information with continuous-variable systems [1]. In order to leverage the power of photonic states, we must be able to reconstruct quantum states with great precision, under the presence of noise, and with an efficiency that unlocks practical implementations. This task is challenging due to the requirement of large Hilbert spaces and unavoidable decoherence and have been approached through different methods [2]. The standard way is to recover the density matrix through iterative Maximum Likelihood Estimation (iMLE), but this method struggles with high dimensional Hilbert spaces [3]. MLE assisted with machine learning and Bayesian inference have arisen as promising options [4–7]. For the scope of this report, I will focus on Bayesian inference to reconstruct posterior samples of parameters through homodyne data measurements of single-photon and squeezed states. I emphasize that this technique has been widely applied previously [6–8], therefore this report does not present a new approach, its purpose is to document and guide on how to implement Bayesian inference for Gaussian and non-Gaussian states.

The main motivation of this project comes from a bigger research on how normalizing flow with machine learning can assist the sampling of posterior distribution more efficiently [9]. Aiming to start from a more fundamental level, I used experimental realizations of actual Gaussian and non-Gaussian optical states to infer parameters through posterior samples, while simultaneously estimating uncertainty and parameter correlations.

Let us start by considering a homodyne dataset $D = \{(x_i, \theta_i)\}_{i=1}^N$ and a parametrized quantum state $\rho(\mathbf{s})$. By introducing the projector operator $\Pi(x_i, \theta_i) = |\psi_i\rangle\langle\psi_i|$,

we can calculate the probability distribution over the quadrature value $\hat{X}_\theta = \hat{X} \cos \theta + \hat{P} \sin \theta$ through the formula [2, 3]

$$p(x_i | \theta_i, \mathbf{s}) = \text{Tr} [\Pi(x_i, \theta_i) \rho(\mathbf{s})]. \quad (1)$$

The statistical nature of measurements in quantum mechanics strongly suggests a link with Bayes' theorem. The posterior probability of the model parameters given a dataset is given by [6–8]

$$p(\mathbf{s} | D) = \frac{p(D | \mathbf{s}) p(\mathbf{s})}{p(D)}, \quad (2)$$

where $p(D|\mathbf{s})$ is the likelihood associated with the measured quadratures, $p(\mathbf{s})$ is the prior information over the parameters, and $p(D)$ is the distribution of the dataset. Eq. (1) provides the likelihood which can be written for the whole dataset and conveniently written in the form

$$\log L(\mathbf{s}) = \sum_{i=1}^N \log p(x_i | \theta_i, \mathbf{s}). \quad (3)$$

Taking into account that $p(D)$ does not depend on the parameters $\mathbf{s}$, it is useful to normalize the posterior $\int p(\mathbf{s} | D) d\mathbf{s} = 1$. Therefore the relevant proportionality is

$$p(\mathbf{s} | D) \propto p(D | \mathbf{s}) p(\mathbf{s}). \quad (4)$$

The posterior distribution is sampled by the Metropolis–Hastings algorithm, which is a subset of the Markov

chain Monte Carlo (MCMC) method for obtaining a sequence of random samples. In addition, MCMC samples the posterior just with the information of relative probabilities, therefore this constant gets canceled out.

The Bayesian parameter estimation (BPE) framework is performed for two simple continuous-variable quantum states generated experimentally and measured through homodyne detection. The first one is the squeezed thermal state

$$\rho_{\text{sq-th}}(\mathbf{s}) = S(r, \phi) \rho_{\text{th}}(n_{\text{th}}) S^\dagger(r, \phi) \quad (5)$$

with $\mathbf{s} = (r, \phi, n_{\text{th}})$. Here r is the squeezing level, ϕ the squeezing angle, and n_{th} represents thermal noise. The second model is the noisy single-photon state

$$\rho(c) = c|0\rangle\langle 0| + (1 - c)|1\rangle\langle 1| \quad (6)$$

with $0 \leq c \leq 1$. I analyze the dependence of the inferred parameters with the number of homodyne measurements. The uncertainties are measure as the width of the 68% credible interval $\Delta_{68} = q_{84} - q_{16}$, which for symmetrical states approximates one standard deviation. q_α is defined by $\int_{-\infty}^{\alpha} p(s | D) ds = \alpha$, so that q_{16} and q_{84} correspond to the 16th and 84th percentiles of the posterior distribution, respectively.

II. CASE STUDIES

I consider four case studies in order to examine the behavior of Bayesian parameter estimation under both controlled and experimental conditions. The first two correspond to a Gaussian squeezed-thermal state, while the latter two consider a non-Gaussian vacuum-single-photon mixture. For each model, I first analyze simulated data, for which the ground-truth parameters are known, and then apply the same inference procedure to experimentally acquired homodyne data. In addition to the final posterior distributions, I study how the inferred parameters and their uncertainties evolve as the number of homodyne measurements is increased.

A. Gaussian State: Squeezed Thermal State

The first model is the squeezed-thermal state introduced above, parametrized by the squeezing strength r_{dB} , the squeezing angle ϕ , and the thermal occupation n_{th} . In this case the quadrature statistics depend explicitly on the homodyne phase, allowing the three parameters to be inferred from the phase-dependent variance of the measured quadratures.

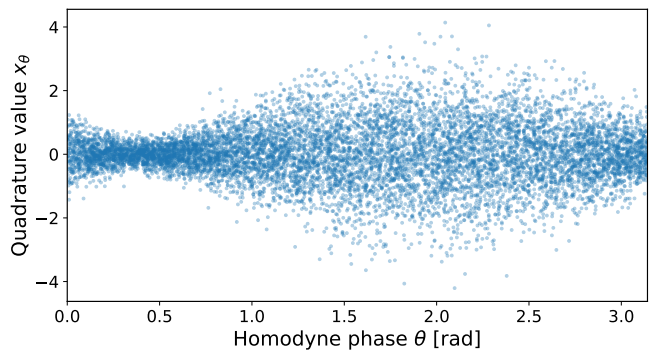


FIG. 1. Simulated homodyne quadrature sequence for the squeezed thermal state as a function of the local-oscillator phase.

1. Simulated Dataset

For the simulated example, I generate homodyne quadrature measurements using the ground-truth parameters $r_{\text{dB}} = 7.0$, $\phi = \pi/4$, and $n_{\text{th}} = 0.10$. Figure 1 shows the resulting quadrature sequence. The phase-dependent width of the distribution directly reflects the squeezed and anti-squeezed quadratures of the state.

The posterior distributions obtained from the simulated dataset are shown in Fig. 2(a). The one-dimensional marginalized distributions are sharply localized around the corresponding ground-truth values. The joint posterior distributions additionally reveal correlations between the inferred parameters. In particular, the squeezing strength and thermal occupation exhibit a visible correlation because both parameters influence the measured quadrature variance.

To examine the statistical behavior of the estimation, I repeat the inference using progressively larger subsets of the same simulated dataset. The results are shown in Fig. 2(b). For small datasets, the posterior estimates exhibit relatively large statistical fluctuations and broad credible intervals. As the number of homodyne measurements increases, the posterior estimates approach the ground-truth parameters and their uncertainties decrease. The credible-interval width $\Delta_{68} = q_{84} - q_{16}$ approximately follows the expected $N^{-1/2}$ scaling.

2. Experimental Dataset

I next apply the same model and inference procedure to an experimentally measured squeezed-state homodyne dataset. In contrast to the simulated case, the true values of the state parameters are not independently known. The purpose of the analysis is therefore not to demonstrate convergence toward a known target, but to determine whether the posterior estimates stabilize as additional experimental measurements are included.

Figure 3 shows the experimental quadrature sequence.

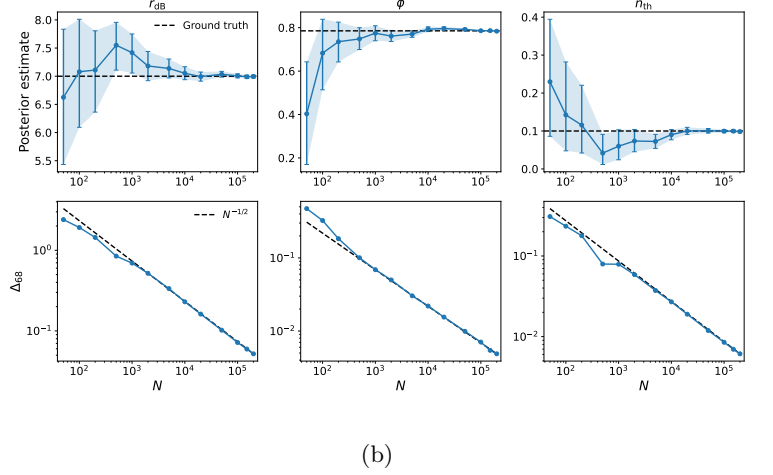
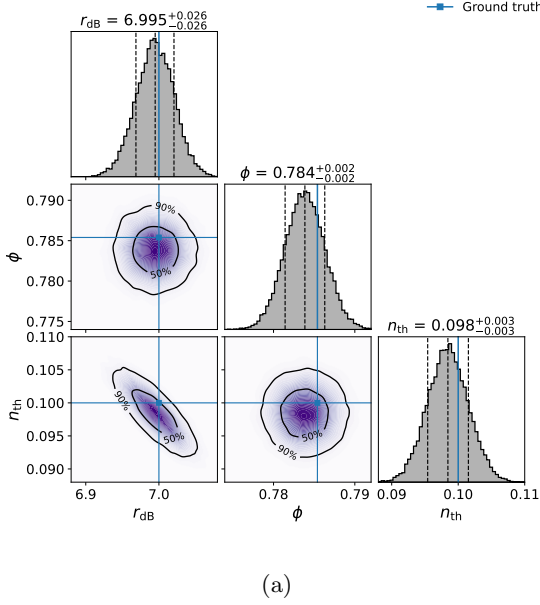


FIG. 2. Bayesian parameter estimation for the simulated squeezed-thermal state. (a) Posterior distributions for r_{dB} , ϕ , and n_{th} . The diagonal panels show the marginalized posterior distributions, while the lower panels show the corresponding joint posterior densities. The blue lines and markers indicate the ground-truth values. Black dashed lines denote the 16th, 50th, and 84th percentiles, while the black contours enclose the indicated posterior probability regions. (b) Bayesian parameter convergence and posterior contraction as the number of homodyne measurements N is increased. The upper panels show the posterior estimates and their central 68% credible intervals, while the lower panels show $\Delta_{68} = q_{84} - q_{16}$ together with the $N^{-1/2}$ reference scaling.

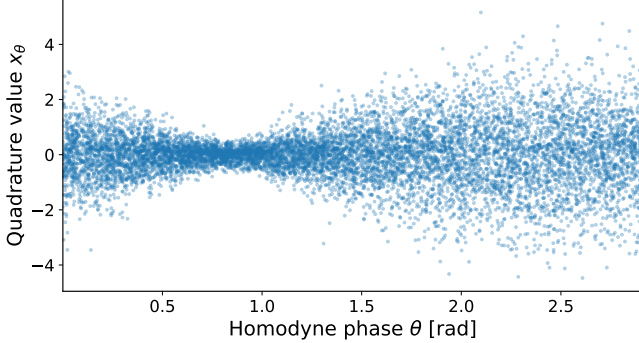


FIG. 3. Experimental homodyne quadrature sequence for the squeezed-thermal state.

The phase-dependent variation of its width is clearly visible and is consistent with the expected behavior of a squeezed optical state.

The inferred posterior distributions and their evolution with increasing dataset size are summarized in Fig. 4. The marginalized posteriors provide estimates of r_{dB} , ϕ , and n_{th} , while the joint posterior distributions reveal correlations between the physical parameters. Since no independent ground-truth values are available, the posterior widths quantify uncertainty within the assumed squeezed-thermal model rather than an error relative to a known state.

As the number of measurements increases, the inferred

parameters become progressively more stable and the posterior credible intervals contract. The $N^{-1/2}$ dependence is included as a reference for the statistical contraction expected in a regular estimation problem. Deviations from this reference may arise in experimental data because of finite-sample effects, experimental noise, calibration uncertainty, or model mismatch.

B. Non-Gaussian State: Noisy Single-Photon State

I next consider the simpler one-parameter non-Gaussian model $\rho(c) = c|0\rangle\langle 0| + (1 - c)|1\rangle\langle 1|$. Unlike the squeezed thermal state, both Fock states appearing in this mixture are rotationally symmetric in phase space. Consequently, the ideal quadrature distribution is independent of the homodyne phase, and the inference is determined by the shape of the measured quadrature distribution.

1. Simulated Dataset

For the simulated example I choose $c = 0.50$, corresponding to an equal mixture of vacuum and single-photon contributions. The resulting homodyne sequence is shown in Fig. 5. Since the state is phase independent, no systematic variation with the homodyne phase is expected. The marginal quadrature distribution shown on

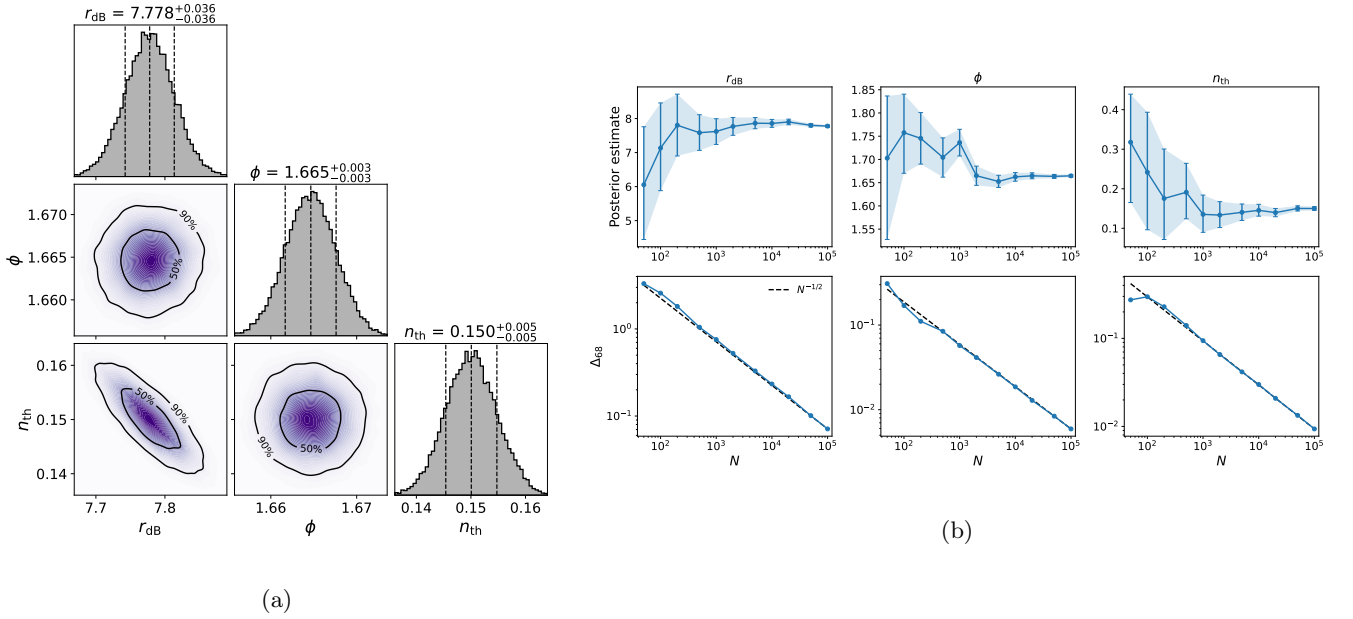


FIG. 4. Bayesian parameter estimation for the experimental squeezed-thermal state. (a) Posterior distributions for r_{dB} , ϕ , and n_{th} . The diagonal panels show the marginalized posterior distributions, while the lower panels show the corresponding joint posterior densities. Black dashed lines indicate the 16th, 50th, and 84th percentiles. (b) Stabilization of the inferred parameters and contraction of their posterior uncertainties with increasing experimental dataset size. The lower panels show Δ_{68} together with the $N^{-1/2}$ reference scaling.

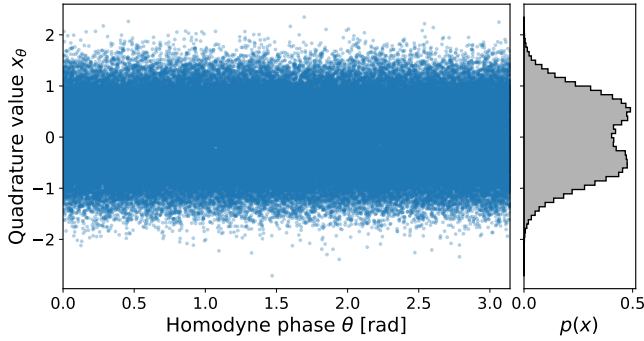


FIG. 5. Simulated homodyne quadrature sequence for the vacuum-single-photon mixture. The marginal quadrature distribution is shown on the right.

the right displays the non-Gaussian structure associated with the single-photon component.

Because only one parameter is inferred, the posterior can be displayed directly as a one-dimensional probability density. Figure 6 shows that the posterior becomes sharply localized close to the true value $c = 0.50$. The dashed lines indicate the central credible interval and posterior median.

The convergence behavior is shown in Fig. 7. For small datasets, the inferred value of c exhibits visible statistical fluctuations, while larger datasets progressively localize the posterior around the true value. The corresponding credible-interval width again follows the expected $N^{-1/2}$

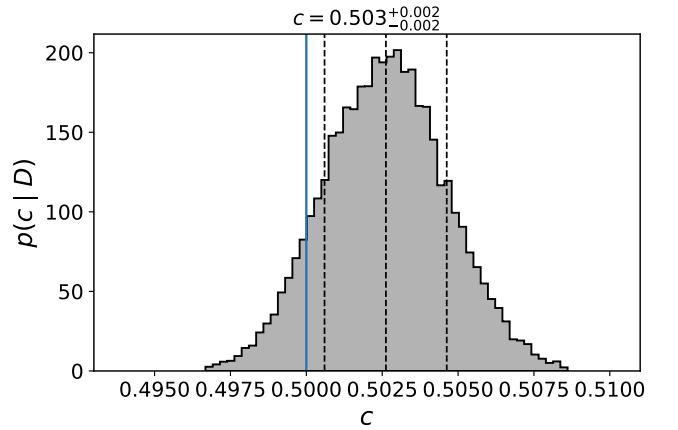


FIG. 6. Posterior distribution of the mixing parameter c for the simulated non-Gaussian state. The blue line indicates the ground-truth value, while the black dashed lines denote the 16th, 50th, and 84th percentiles.

contraction closely.

2. Experimental Dataset

Finally, I apply the same single-parameter model to experimental single-photon homodyne data. The measured quadrature sequence and its marginal distribution are shown in Fig. 8. The marginal distribution retains a

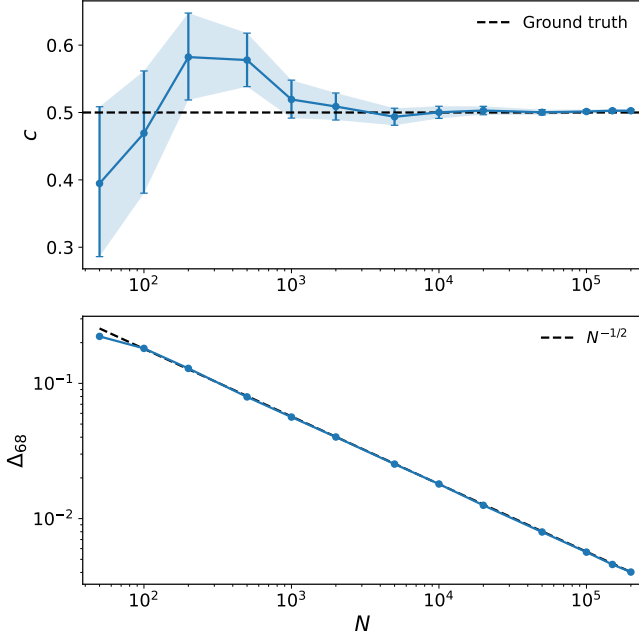


FIG. 7. Bayesian convergence of the inferred mixing parameter c and contraction of its posterior uncertainty with increasing number of homodyne measurements. The lower panel shows Δ_{68} together with the $N^{-1/2}$ reference scaling.

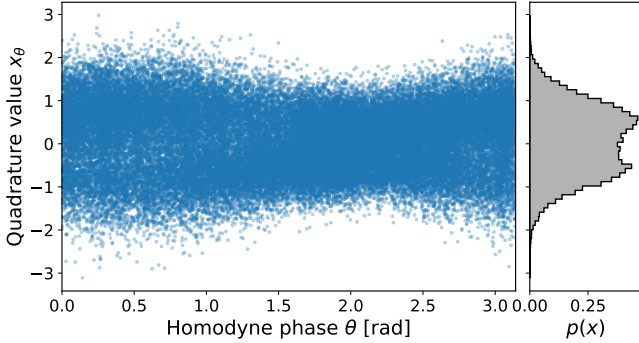


FIG. 8. Experimental homodyne quadrature sequence for the non-Gaussian optical state. The marginal quadrature distribution is shown on the right.

clear non-Gaussian, double-peaked structure associated with a significant single-photon contribution. At the same time, the phase-resolved quadrature sequence exhibits a small variation of its width with the homodyne phase. Such phase dependence is not contained in the simple vacuum–single-photon mixture and therefore indicates that the model should be regarded as an effective approximation to the experimental state rather than a complete physical description.

The posterior distribution of c is shown in Fig. 9. The posterior is well localized, indicating that the effective mixing parameter is strongly constrained by the measured quadrature distribution. Importantly, a narrow

posterior does not by itself demonstrate that the assumed

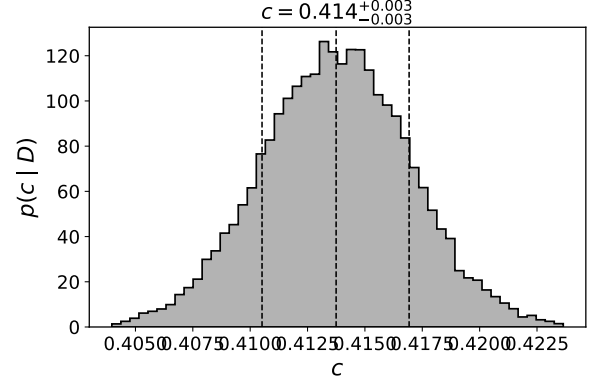


FIG. 9. Posterior distribution of the mixing parameter c inferred from the experimental non-Gaussian homodyne data. Black dashed lines denote the 16th, 50th, and 84th percentiles.

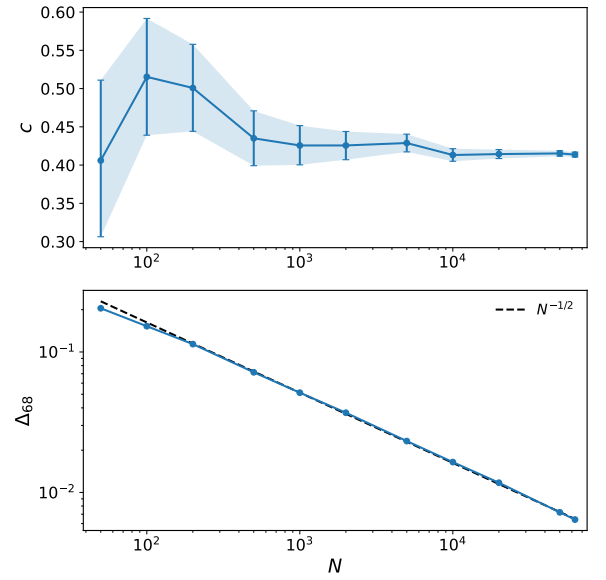


FIG. 10. Stabilization of the inferred mixing parameter c and contraction of the central 68% credible interval with increasing experimental dataset size. The dashed line indicates the reference scaling $N^{-1/2}$.

model provides a complete description of the experimental state; it quantifies the uncertainty in c conditional on that model.

The evolution of the inferred parameter and posterior width is presented in Fig. 10. As the number of measurements increases, the estimate of c becomes increasingly stable and the credible interval contracts. The comparison with the $N^{-1/2}$ reference provides a simple diagnostic of the statistical scaling of the inference. For the experimental case, the primary quantity of interest is the stabilization of the estimate, since no independent ground-truth value of c is available.

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